

REFNO: 074904

Roll No. _____

B.Sc. (Hons.) Semester II Examination 2023-24

Subject: Mathematics

Paper No.: MTB-202 - Statics & Dynamics

Time : 3 Hours]

[Maximum Marks : 70

[Write your Roll No at the top immediately on the receipt of this question paper.]

Answer five questions including Question No. 1 which is compulsory.

The figures in the right hand margin indicate marks.

1. (a) Obtain a force or a couple which is equivalent to a system consists of four forces with components $(2, 3)$, $(3, -1)$, $(-4, -4)$ and $(-1, 2)$ acting on a rigid body at the points $(0, 1)$, $(1, -1)$, $(2, 3)$ and $(-1, -2)$, respectively. **2**
- (b) Find out the work done by the thrust at the ends of a rod when displacement alters its length. **2**
- (c) Prove that a system of coplanar forces acting at different points of a rigid body will be in equilibrium if the algebraic sums of the moments of all the forces with respect to each of three non-collinear points are zero. **2**
- (d) If a particle of mass m moving along any path has the linear velocity v at a distance r from the origin, then obtain the angular velocity ω of the particle. **2**
- (e) Choose the statements which are not the properties of a simple harmonic motion of a particle from the list given below:
 - (i) The acceleration of the particle is always directed towards a fixed line.
 - (ii) While in motion the particle will maintain a constant distance from a fixed point.
 - (iii) The linear velocity of the particle will be constant at all points.
 - (iv) The particle will always move in a straight line.**2**
- (f) If the velocity components of a moving particle in the radial and transverse directions are λr and $\mu \theta^2$ (λ and μ are constants), then find the polar equation of the path. **2**
- (g) State the mathematical form of Kepler's third law of planetary motion and prove it. **2**
- (h) If a particle describes a cardioid, $ap^2 = r^3$ under a force which is always directed towards the pole, then find the law of force. **2**
- (i) Is the areal velocity of a particle moving under a central force always proportional to its angular momentum per unit mass? Give reasons for your answer. **2**
- (j) A particle of mass 3 unit moving upward along the inner surface of a smooth vertical circular path of radius 5 unit with initial velocity 15 unit. Find out the reaction of the ring on the particle at the position having angle 60° when $g = 9.8$ unit. **2**
- (k) A frame has a linear motion with acceleration. Is the frame a Newtonian frame of reference? Give reasons for your answer. **2**

(2)

2. (a) Show that a system of coplanar forces acting at different points of a rigid body can be reduced to a single force through a given point and a couple. **6**
- (b) A system of forces in the plane of a triangle ABC is equivalent to a single force at A , acting along the internal bisector of the angle BAC , and a couple G_1 . If the moment about B and C are G_2 and G_3 respectively, show that $(b+c)G_1 = bG_2 + cG_3$, where $b = CA$ and $c = AB$. **6**
3. (a) State and prove the principle of virtual work for a system of coplanar forces acting at different points of a rigid body. **6**
- (b) A heavy elastic string, whose natural length is $2\pi a$, is placed round a smooth cone whose axis is vertical and whose semi-vertical angle is θ . If W be the weight and λ be the modulus of elasticity of the string, then prove that it will be in equilibrium when it forms a circle of radius $a \left(1 + \frac{W}{2\pi\lambda} \cot \theta\right)$. **6**
4. (a) Derive the tangential and normal components of acceleration of a particle moving in a plane curve. Also, write down the expressions of radial and transverse components of acceleration. **6**
- (b) A point moves in a curve so that its tangential and normal accelerations are equal and the tangent rotates with constant angular velocity. Show that the intrinsic equation of the path is of the form $s = Ae^\psi + B$; A and B are constants. **6**
5. (a) Show that in a SHM the average speed and the average acceleration (in magnitude) can be obtained by multiplying their maximum values by $2/\pi$. **6**
- (b) Discuss the motion of a particle in a frame of reference which is rotating with constant angular velocity about a fixed frame and obtain the generated fictitious forces. **6**
6. (a) If the central orbit is $r^n = a^n \cos n\theta$ under a force towards the pole then find the law of force and the velocity at any position. **6**
- (b) A particle under central acceleration $\mu/(\text{distance})^2$ is projected with velocity V at a distance R . Show that its path is a rectangular hyperbola if the angle of projection is $\sin^{-1} \left\{ \frac{\mu}{V} / \sqrt{V^2 R^2 - 2\mu R} \right\}$. **6**
7. (a) A particle is free to move on a smooth vertical circular wire of radius a . If it is projected from the lowest point with velocity $\sqrt{4ga}$, then show that the reaction between the particle and the wire will be zero after a time $\sqrt{a/g} \log(\sqrt{5} + \sqrt{6})$, g being the acceleration due to gravity. **6**
- (b) Two particles are let drop from the cusp of a cycloid down the curve at an interval of time t . Prove that they will meet in a time $2\pi\sqrt{a/g} + t/2$. **6**

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